Anwesan Pal

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Research Interest - To pursue challenging research questions in the domain of Computer Vision and Machine Learning with focus on semantic scene understanding and embodied-AI agents.

EDUCATION

Academic Qualifications

Doctor of Philosophy (Ph.D.) — **La Jolla, USA**
Computer Science and Engineering, UC San Diego — *2019 - Present*
Advisor: Dr. Henrik I. Christensen

Master of Science (M.S.) — **La Jolla, USA**
Electrical and Computer Engineering, UC San Diego — *2017 - 2019*
Advisors: Dr. Henrik I. Christensen & Dr. Nuno Vasconcelos
Title: Learning Application-Oriented Classifiers for Multi-frame Visual Recognition

Bachelor of Engineering (B.E.) — **Shibpur, India**
Electrical Engineering, IIEST Shibpur — *2013 - 2017*
Advisors: Dr. Aparajita Sengupta & Dr. Suvarun Dalapati
Title: Fabrication of Prototype Apparatus for Control Systems Laboratory Experiment
Awarded highest honor for undergraduate thesis

PUBLICATIONS

Accepted

- **A. Pal***, Y. Qiu*, and H. I. Christensen, "Learning hierarchical relationships for object-goal navigation," in Conference on Robot Learning (CoRL), Cambridge, USA, Nov. 2020. (acceptance rate - 34%)
- **A. Pal**, S. Mondal, and H. I. Christensen, ""Looking at the right stuff" - Guided semantic-gaze for autonomous driving," in IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, USA, Jun. 2020. (acceptance rate - 22.08%)
- **A. Pal**, C. Nieto-Granda, and H. I. Christensen, "DEDUCE: Diverse scEne Detection methods in Unseen Challenging Environments," in IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Macau, China, Nov. 2019.

Ongoing

- H. Xiang, **A. Pal**, Y. Qiu, Q. Li, and H. I. Christensen, "Knowledge-based trajectory prediction with attentive intent"

PROJECTS

Research Oriented

Visual Semantic Navigation — **UC San Diego**
Supervisor - Prof. Henrik I. Christensen — *Jan 2020 - Present*

- Proposed an object-goal navigation algorithm utilizing hierarchical object relationships
- Published paper as joint first author at **CoRL 2020**

Saliency models for Autonomous Driving — **UC San Diego**
Supervisor - Prof. Henrik I. Christensen — *Jun 2019 - Nov 2019*

- Developed a framework for predicting driver's focus of attention from surrounding semantic context
- Published paper as first author at **CVPR 2020**

*Equal Contribution

Place Categorization for semantic mapping **UC San Diego**
Supervisor - Prof. Henrik I. Christensen *Nov 2018 - Mar 2019*

- Proposed an object-based visual recognition model for robot localization and mapping in unknown scenes
- Published paper as first author at **IROS 2019**

Multi-frame Visual Recognition **UC San Diego**
Supervisor - Prof. Nuno Vasconcelos *Jan 2018 - Aug 2018*

- Studied *bias* of action-recognition datasets towards different *representations* and *algorithms*
- Submitted paper as first author at **CVPR 2019**

Academic Oriented

Object Detection in the Autonomous Driving Scenario **UC San Diego**
Course - ECE 285 ***Grade: A+*** *Mar 2019 - Jun 2019*

- Evaluated the performance of three state-of-the-art multi-object detectors on a real autonomous-driven vehicle in the UCSD campus
- Proposed an algorithm for predicting sub-classes of a car object for fine-grained analysis through an end-to-end deep network

AgeGAN: Age Progression/Regression and Estimation **UC San Diego**
Course - COGS 260 ***Grade: A*** *Mar 2018 - Jun 2018*

- Generated future and past estimates of a face from given input image
- Estimated accurate age-group from a given input image as an unsupervised learning problem

Optical Flow-based Video Classification with Visual Attention **UC San Diego**
Course - ECE 271B ***Grade: A*** *Jan 2018 - Mar 2018*

- Classified human gestures from video sequence with higher accuracy compared to state-of-the-art methods
- Applied Visual Attention module to Flow-based Recurrent Neural Networks

Fabrication of prototype apparatus for Control Systems Laboratory Experiment **IIEST Shibpur**
Course - EE 854 ***Grade: A*** *Jul 2016 - May 2017*

- Designed a *Linear State Variable Feedback (LSVF)* Controller for low-gain applications
- Fabricated the prototype in a dotted Veroboard
- Awarded highest honor for undergraduate thesis

WORK EXPERIENCE

Research Oriented

Graduate Research Assistant **La Jolla, USA**
UC San Diego *Sep 2018 - Present*

- Multi-agent perimeter defense and hot-spot identification
- Target-driven visual navigation for object goal tasks
- Learning-based saliency prediction of focus of attention (FoA) for autonomous vehicles
- Object-based semantic mapping and place categorization

Summer Research Fellow **La Jolla, USA**
UC San Diego *Jun 2018 - Aug 2018*

- Proposed 6 types of *representation* bias for action recognition
- Characterized bias both at dataset and individual action-class level
- Assess strengths and weaknesses of existing action-recognition datasets towards algorithm bias

Research Intern **Paris, France**
Paris VIII University *May 2016 - Jul 2016*

- Defined behavioral algorithms for interaction among multiple-agents of NAO Humanoid
- Simulated a soccer game using CABSL/XABSL language for NAO agents
- Competed as part of Paris VIII University team in RoboCup 2017 tournament

Research Intern **Goa, India**
BITS Pilani Goa Campus *May 2015 - Jun 2015*

- Studied about embedded micro-controllers in small-scale robots
- Implemented motion control for omni-directional Firebird V to follow series of concentric circular patterns

Teaching Oriented

Graduate Teaching Assistant **UC San Diego**
Course - PHYS 120 *Mar 2018 - Jun 2018*

- Responsibilities: Proctored laboratory sessions, graded assignments and occasionally presented lectures
- Class strength: $\sim$ 80 upper-division undergraduate students

Graduate Teaching Assistant **UC San Diego**
Course - ECE 276A *Sep 2017 - Dec 2017*

- Responsibilities: Held discussion sessions and graded assignments
- Class strength: $\sim$ 120 graduate students

AWARDS AND ACHIEVEMENTS

- Awarded Graduate Fellowship for Doctoral study by the CSE Department at UC San Diego
- Awarded Summer Research Fellowship by the ECE Department at UC San Diego to conduct research in a laboratory environment
- Only student in the batch to be a Teaching Assistant for a graduate-level course in the ECE Department in the first quarter in UC San Diego
- Attained perfect CGPA of 10.0 (out of 10) in class X. Received Scholarship and Letter of Recognition from Minister of Education, Govt. of India
- Stood 2nd in *WIRED-IN* – the Electrical/Electronics Design contest held in INSTRUO 2015, technical festival of IIEST, Shibpur

REVIEWER SERVICES

- IEEE International Conference on Robotics and Automation 2021
- IEEE/RSJ International Conference on Intelligent Robots and Systems 2020
- IEEE/CVF Conference on Computer Vision and Pattern Recognition 2019

TEST SCORES

Graduate Record Examination (GRE)

Verbal - 151, Quantitative - 168, Analytical Writing - 4.5

Test Of English as Foreign Language (TOEFL)

Reading - 26, Listening - 26, Speaking - 24, Writing - 27

English Language Certification Exam (ELCE) - conducted by UC San Diego

Pass with a grade of 4/4

SKILLS

- **Programming Languages:** Proficient in: Python, Matlab, C/C++
- **Deep Learning Frameworks & Tools:** PyTorch, TensorFlow, Keras, Caffe, Simulink
- **Operating Systems:** Linux, Windows, MacOS, Robot Operating System (ROS)
- **Languages:** Proficient in: English, Hindi. Native: Bengali

INTERESTS

- Teaching Physics, Mathematics and Computer Science to high school and undergraduate students.

- Solving puzzles, reading science-fiction, watching detective sitcoms.
- Listening to light music, playing video-games at off-time.

REFERENCES

Henrik I. Christensen
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